

Prefill-Context Execution Pass-Rate Eval

V2 Design Document

Incorporates V1 failure mode audit + reviewer feedback

Version 2.0 | 2026-05-14

Model: Claude Sonnet 4.6 via Max Plan | Platform: GB10 ARM64

1. V1 Post-Mortem: Why the Results Were Wrong

V1 reported a 43.3% pass rate (26/60). Independent review identified that this number was contaminated by three categories of infrastructure bugs that inflated the failure count. A failure mode audit of all 41 V1 FAIL trials revealed:

Failure Mode	Count	%	Is Real Failure?
PATH_ERROR (correct answer)	11	26.8%	NO - false negative
TIMEOUT (900s)	9	22.0%	YES - model too slow
FORMAT_MISMATCH	9	22.0%	PARTIAL - GT/test misalign
PATH_ERROR (no answer)	6	14.6%	UNCLEAR
LOGIC_ERROR	6	14.6%	YES - genuine failure

Key findings from the audit:

1. Only 6/41 failures (14.6%) were genuine LOGIC_ERRORS where the model produced wrong answers. The majority were infrastructure artifacts.
2. 11 failures were false negatives: the model computed the correct answer but wrote it to a path the rewritten tests could not find (e.g., output/mass_report.json instead of workspace root).
3. ALL 9 azure-bgp failures were FORMAT_MISMATCH (test ERRORS, not FAILURES). The GT SKILL.md guided the model to produce output in a format the test suite did not expect. This is a SKILL.md-test alignment defect, not a model competence issue.
4. V1's corrected pass rate (after accounting for false negatives) was ~60%, not 43%.

1.1 Invalidated V1 Claims

- 'GT-only is worst (20%)' -- partially an artifact of azure-bgp format mismatch. The SKILL.md taught a format the test didn't accept.
- 'Noise helps execution (43% > 40%)' -- the 3pp difference was within statistical noise at N=5, and contaminated by false negatives.
- 'Hard noise is best (50%)' -- may be a length/attention artifact rather than a semantic diversity effect, given the GT/stub asymmetry.

2. V2 Pipeline Changes

V2 implements four fixes to address the measurement errors identified in V1, plus two additional baseline conditions from reviewer feedback.

2.1 Fix: Output Path Reconciliation

Problem: The model sometimes writes correct output to alternative paths (output/file.json, /root/file.json) instead of the workspace root where rewritten tests expect it.

Fix: After the agent exits and before running tests, a new `_reconcile_output_paths()` function copies files from common alternative locations (output/, root/, app/, app/output/) to the workspace root.

```
def _reconcile_output_paths(work: Path) -> None:
    for sub in ('output', 'root', 'app', 'app/output'):
        d = work / sub
        if d.is_dir():
            for f in d.rglob('*'):
                if f.is_file():
                    dst = work / f.relative_to(d)
                    if not dst.exists():
                        shutil.copy2(f, dst)
```

2.2 Fix: System Prompt File (ARG_MAX)

Problem: Large system prompts (>128KB for sz=50 hard noise) exceeded Linux ARG_MAX when passed as --system-prompt command-line argument, causing 6 trials to crash silently.

Fix: Write system prompt to a temporary file and use --system-prompt-file flag.

2.3 Fix: Dockerfile Filter Regex

Problem: The regex `/tex/` matched 'bibtexparser' in citation-check's Dockerfile, causing all 12 citation-check trials to be skipped as 'heavy Dockerfile'.

Fix: Changed regex from 'tex' to 'texlive' to avoid false matches.

2.4 Fix: Sequential Container Execution

Problem: Python asyncio with multiple coroutines caused the process to hang after 3-16 trials when running inside Docker containers.

Fix: --max-trials 1 flag ensures each container runs exactly one trial then exits. A bash loop (run_sequential.sh) manages the trial queue with resume support.

2.5 New: Baseline Conditions

Baseline	Description	Purpose
noskill	No SKILL.md in system prompt	Is skill injection useful at all?
gtonly	Only GT skill(s), zero noise	Is noise harmful or beneficial?

3. V2 Experimental Design

3.1 Main Experiment (unchanged from V1)

Variable	Levels	Description
Pool Size	5, 10, 20, 50	Total candidates (GT + noise)
Noise Mode	random, hard, easy	Distractor sampling strategy
Seed	0	Deterministic pool construction

5 tasks x 4 sizes x 3 noise modes = 60 trials (main experiment).

3.2 Baselines (new in V2)

Two additional conditions run per task (5 trials each, 10 total):

- noskill: Model receives only the task instruction, no SKILL.md content.
- gtonly: Model receives only the GT skill(s), no noise distractors.

These establish the lower bound (can the model solve it without help?) and the upper bound (does the correct skill actually help?).

3.3 Failure Mode Classification (new in V2)

Every FAIL trial is automatically classified into one of five categories:

Mode	Detection Rule	Interpretation
TIMEOUT	agent_wall_s >= 900	Model too slow / stuck in loop
PATH_ERROR_TRUE_PATH	NotFoundError + correct answer in st	FALSE NEGATIVE - fix needed
PATH_ERROR_FileNotFoundError	NotFoundError, no correct answer detected	Ambiguous
FORMAT_MISMATCH	Test ERRORS (not FAILURES)	SKILL.md/test misalignment
LOGIC_ERROR	Test FAILURES (assertions failed)	Genuine model error

This classification runs in the cron monitoring job every 10 minutes. If 3 consecutive failures share the same non-LOGIC mode, it triggers a bug alert for investigation.

4. Architecture (V1 -> V2 Delta)

4.1 Trial Execution Flow

V1 flow:	V2 flow:
setup_workspace()	setup_workspace()
patch_tests()	patch_tests()
patch_instruction()	patch_instruction()
build_system_prompt()	build_system_prompt()
	write system_prompt to temp file [NEW]
claude -p --system-prompt ...	claude -p --system-prompt-file .. [CHANGED]
	_reconcile_output_paths() [NEW]
run_tests()	run_tests()
	classify_failure_mode() [NEW]
emit JSONL	emit JSONL

4.2 Orchestration

V1: docker compose up

- > python run_trial.py --concurrency 2
- > asyncio creates ALL coroutines
- > HANGS after 3-16 trials

V2: bash run_sequential.sh

- > while loop
 - > docker run --rm (fresh container per trial)
 - > python run_trial.py --max-trials 1 --resume
 - > exits after 1 trial
 - > check progress, stall detection (3-strike abort)
- > aggregate results

4.3 Monitoring (cron job)

A 10-minute cron job automatically:

1. Checks container status
2. Reports PASS/FAIL/total progress
3. Classifies failure modes for all FAIL trials
4. Detects bug patterns (3+ consecutive same-type failures)
5. Compares against V1 results

5. Known Limitations (Honest Assessment)

5.1 GT/Noise Content Asymmetry (NOT fixed in V2)

GT skills receive full SKILL.md (often 1000+ words). Noise skills receive 50-word corpus stubs. A sufficiently capable model can identify GT by content length alone, without semantic understanding.

This confound affects noise-mode comparisons (hard vs easy vs random) but does NOT affect the baseline comparison (noskill vs gtonly vs prefill), because:

- noskill has no SKILL.md at all
- gtonly has one full document with nothing to compare against

Fix for future work: symmetric content ablation where all candidates receive full SKILL.md content.

5.2 azure-bgp FORMAT_MISMATCH (NOT fixed in V2)

The azure-bgp GT SKILL.md guides the model to produce output in a structure that the test suite's parametrized assertions cannot parse. This is a SkillsBench data quality issue, not a pipeline bug.

V2 does NOT modify task data (SKILL.md, tests, instructions). The FORMAT_MISMATCH failures for azure-bgp will persist. They should be excluded from noise-mode analysis or treated as a separate data-quality finding.

5.3 Sample Size (N=5 per task)

The smoke run uses 5 tasks. Individual task-level or cell-level comparisons are not statistically significant. The smoke run's purpose is to validate the pipeline and identify measurement bugs, not to draw conclusions.

The full run (84 tasks x 12 conditions = 1,008 trials) is required for paper-grade statistical power.

5.4 Pool Size / Context Length Confound

Pool size and system prompt length are perfectly correlated:

- sz=5: ~5K chars
- sz=50: ~35K chars

Any pool-size effect may be driven by context length, not pool composition. Isolating this requires a 'random long text' baseline with matched token count but no procedural content.

6. V2 Early Results (In Progress)

V2 re-run is in progress. Early results show dramatic improvement from the path reconciliation fix:

Metric	V1	V2 (partial)
3d-scan-calc pass rate	10/12 (83%)	12/12 (100%)
Overall pass rate (so far)	26/60 (43%)	15/16 (94%)
PATH_ERROR_TRUE_PASS count	11	0
False negative rate	26.8% of failures	0%

The two 3d-scan-calc trials that failed in V1 due to path errors now pass in V2. The single V2 failure so far is a genuine TIMEOUT (adaptive-cruise-control sz5 random, 900s). No PATH_ERROR or FORMAT_MISMATCH bugs detected.

V2 run is ongoing. Final results will be available after all 60 main trials + 10 baseline trials complete (~4-6 hours from start).

7. Reviewer Concern Response Plan

Concern	Priority	Status in V2	Evidence
Path rewriting false negatives	P0	FIXED	<code>_reconcile_output_paths()</code>
Failure mode classification	P0	FIXED	Auto-classification in cron
GT/test format alignment audit	P0	DOCUMENTED	azure-bgp identified, not fixed
Missing noskill/gtonly baselines	P1	IMPLEMENTED	Code ready, V1 data collected
GT/noise content asymmetry	P1	NOT FIXED	Needs symmetric ablation exp
N=5 sample size	P2	ACKNOWLEDGED	Smoke run only; full grid needed
Pool size / context length	P2	NOT FIXED	Needs random-long-text baseline

7.1 What V2 CAN Claim

- The V1 pass rate was artificially depressed by infrastructure bugs (path rewriting, ARG_MAX, Dockerfile regex).
- After fixing measurement errors, the true pass rate is significantly higher than V1's 43%.
- PATH_ERROR_TRUE_PASS (false negatives) have been eliminated.
- Failure mode classification enables clean separation of model failures from infrastructure failures.

7.2 What V2 CANNOT Claim (yet)

- 'Noise helps execution' -- requires symmetric content ablation and $N \gg 5$.
- 'Hard noise is best' -- confounded by GT/stub length asymmetry.
- 'Larger pools help' -- confounded by context length.
- 'GT-only hurts' -- partially driven by azure-bgp format defect.

8. Recommended Experiments (Post-V2)

Experiment	Purpose	Trials	Priority
Symmetric full-doc ablation	Isolate length-matching confound	~250	P1
Random long-text baseline	Isolate context-length effect	~60	P1
Multi-seed replication (0,1,2)	Confidence intervals	~180	P1
Full 84-task grid	Statistical power	1,008	P2
Failure mode deep-dive	Classify FAIL stdout	0 (analysis)	P0
azure-bgp SKILL.md fix	Eliminate format mismatch	~12	P1

9. File Reference (V2 Additions)

File	V1/V2	Purpose
run_trial.py	MODIFIED	_reconcile_output_paths, --system-prompt-file, --max-trials, noskill/gtonly modes
pool_builder_v3.py	MODIFIED	noskill + gtonly noise modes
run_sequential.sh	NEW	One-trial-per-container bash orchestrator
run_baselines.sh	NEW	noskill + gtonly baseline runner
generate_report.py	NEW	Analysis report PDF generator
generate_designdoc_v2.py	NEW	This document
results/sb_exec.jsonl	V2 DATA	60-trial main experiment (in progress)
results/sb_baselines.jsonl	V2 DATA	10-trial baselines
results/sb_exec_v1.jsonl	ARCHIVED	V1 main experiment (43.3%)
results/sb_baselines_v1.jsonl	ARCHIVED	V1 baselines
results/analysis_report.pdf	V1 REPORT	V1 analysis (superseded)